

## 4.6 Relational Schema Development

ریلیشنل اسکیم ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔

CS XI

Chapter 4: Data and Analysis

Lecture 08 of 15

## 4.6 Relational Schema Development

### LEARN IT

A **relational schema** is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate. You write it *before* clicking around in Access.

It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships. Benefits: less redundancy, integrity planned in advance, easier to grow later.

### WORKED EXAMPLE

Written schema for a tiny college:

```
Department (DeptID PK, DeptName)
Student    (RollNo PK, Name, Class, DeptID FK → Department)
Course     (CourseID PK, Title, Credit, DeptID FK → Department)
```

### REMEMBER IT

Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

### اردو میں سمجھیں

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### CHECK YOURSELF

1. Name two things a schema must show.
2. Give one reason to write a schema before opening Access.

Answers: 1. Tables/fields and keys/relationships. 2. It reduces redundancy and mistakes (any valid benefit).

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Next lecture: 4.7 Case Study — Library Management System